

| Question |     |      |  | Marking details                                                                                                                                                  | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) |      |  | <b>D</b> (1)<br><br>no change in appearance and no temperature change (when mixed) (1)<br><br>do <b>not</b> credit reason if letter other than <b>D</b> is given | 2               |     |     | 2     |       | 2    |
|          | (b) | (i)  |  | limewater turns milky/cloudy                                                                                                                                     | 1               |     |     | 1     |       | 1    |
|          |     | (ii) |  | $\text{Na}_2\text{CO}_3$                                                                                                                                         |                 | 1   |     | 1     |       |      |
|          | (c) | (i)  |  | <b>C</b>                                                                                                                                                         |                 |     | 1   | 1     |       | 1    |
|          |     | (ii) |  | orange/yellow                                                                                                                                                    | 1               |     |     | 1     |       | 1    |
|          |     |      |  | Question 4 total                                                                                                                                                 | 4               | 1   | 1   | 6     | 0     | 5    |